

Part 1: Literature Survey and Non-Biomedical Field Selection

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Introduction to the GAPMAP Framework

The paper "GAPMAP: Mapping Scientific Knowledge Gaps in Biomedical Literature Using Large Language Models," authored by Nourah M. Salem, Elizabeth White, Michael Bada, and Lawrence Hunter, presents a novel approach to identifying gaps in scientific literature using large language models (LLMs). It was submitted to the 39th Conference on Neural Information Processing Systems (NeurIPS 2025) Workshop on AI for Science. The researchers recognized that the overwhelming number of scientific publications makes it challenging for researchers to identify critical unanswered questions, what they refer to as "known unknowns", that are vital for advancing knowledge. GAPMAP aims to address this issue by utilizing LLMs to detect two types of knowledge gaps: explicit and implicit.

Explicit vs. Implicit Knowledge Gaps

Explicit knowledge gaps are clearly stated deficiencies in knowledge, often highlighted by phrases such as "remains unknown," "further research is needed," or "no randomized controlled trial has evaluated." For instance, an explicit knowledge gap might be: "It remains unknown whether long-term GLP-1 agonist therapy improves renal outcomes in non-diabetic CKD." This statement directly points to a lack of evidence.

On the other hand, implicit knowledge gaps are not overtly expressed but can be inferred from the surrounding context. Identifying these gaps requires more nuanced reasoning and might involve:

1. Claims that don't connect logically
2. Generalization gaps where research findings are limited in application
3. Unresolved conflicting evidence

For example, if one study states, "Compound E improves biomarker F in mice," while another says, "Biomarker F correlates poorly with clinical outcomes in humans," you could infer an

implicit gap: "It is unknown whether E improves patient outcomes." This indicates a discrepancy between experimental and clinical evidence.

To address implicit gaps, GAPMAP employs a method called TABI (Toulmin-Abductive Bucketed Inference). This structured reasoning technique, based on the Toulmin Argument Model, treats gap detection as a form of abductive reasoning. The LLM generates a Claim (the inferred knowledge gap), supported by Grounds (the premises) and a Warrant (the logic that links them). The claims are then assessed for their likelihood, allowing for validation and benchmarking against established standards.

The researchers tested GAPMAP using both open-weight LLMs (like Llama and Gemma 2) and closed-weight models (like OpenAI) across four datasets. One dataset focused on explicit gaps, while another was manually evaluated for implicit gaps. Results showed strong performance, especially with larger models, though there were challenges related to adapting to specific fields and confirming results through human assessment.

While GAPMAP's primary focus is on biomedical literature, the team aims to extend its application to a non-biomedical area to explore its broader relevance. They have chosen cognitive neuroscience, specifically the subfield of visual object recognition, which intersects psychology, neuroscience, and artificial intelligence. This area is crucial for understanding human perception, improving AI vision systems, and addressing clinical issues like neurodevelopmental disorders.

Selected Field and Dataset: Cognitive Neuroscience - Visual Object Recognition

Cognitive neuroscience examines how the brain processes information to form perceptions, memories, and behaviors. Within this realm, visual object recognition investigates how humans identify and categorize objects in their visual surroundings. This fundamental cognitive process involves specific brain regions, particularly the inferior temporal (IT) cortex. Despite advancements in neuroimaging and behavioral studies, many key aspects of object representation in real-world scenarios remain unexplored, making it an ideal candidate for gap mapping.

For this project, we will utilize the THINGS initiative as our dataset. THINGS is a large, multimodal resource aimed at studying how the brain represents objects in behavior. It was developed as part of a collaborative global effort and includes:

- **Image Database:** Over 26,107 high-quality images representing 1,854 distinct object concepts, encompassing both natural items (like animals and fruits) and man-made objects (like vehicles and furniture) to provide a comprehensive sample.
- **Behavioral Data:** More than 4.7 million similarity judgments that create a 66-dimensional embedding, capturing how humans perceive similarities between objects based on various dimensions such as animacy, size, and manipulability.
- **Neuroimaging Data:** Detailed functional MRI (fMRI) and magnetoencephalography (MEG) data collected from participants while they viewed numerous images. For instance, the THINGS-fMRI dataset focuses on the brain responses of multiple subjects as they engage with these images.

Selected Field and Dataset: Behavioral Economics; Decision-Making, Social Preferences, and Behavioral Interventions

Behavioral economics combines insights from psychology with economic models to help explain why people sometimes make choices that deviate from rational decision-making. This area of study looks at various biases, simple mental shortcuts (heuristics), inconsistencies in how we value the future, as well as concepts like reciprocity, trust, nudges (subtle prompts to influence behavior), prosocial behavior, and decisions related to the environment.

Key areas of research in behavioral economics include experimental studies focused on negotiations, cooperation games (such as dictator and ultimatum games and public goods), financial planning, sustainable behavior, and interventions in policy. Despite a wealth of evidence from both laboratory and field studies, there remain significant gaps in our understanding. These gaps can be explicit, such as calls for more research on how interventions can be scaled or their long-term effects and implicit, like inconsistencies between people's attitudes and behaviors, discrepancies between lab findings and real-world applications, unintended consequences of nudges, and influences of cultural or gender differences on trust and reciprocity.

For this project, we will use Volume 119 (December 2025) of the *Journal of Behavioral and Experimental Economics* as our main source of literature and dataset. This volume includes a variety of experimental articles, covering lab experiments, field studies, hybrid approaches, and surveys that examine contemporary topics, including:

- Reducing cognitive inconsistencies in sustainable online shopping [Eßer et al., 2025].
- The unintended consequences of nudges for shared mobility options, like e-scooters [Ocean & Woodman, 2025].
- Challenges in leadership messaging that aims to encourage climate action among producers [Yokoo et al., 2025].
- Effectiveness of financial education programs and their impacts [Llamosas García & Mazas Pérez-Oleaga, 2025].
- Evidence of buyer-side gender discrimination in bargaining from seed sales in Uganda [Van Campenhout & Nabwire, 2025].
- Effects of social distancing on trust and solidarity [Filippin & Pace, 2025].
- The gap between attitudes and actual pro-environmental behaviors [Zíka et al., 2025].
- Moving towards greater openness and reproducibility in behavioral and experimental economic research [Rahal, 2025].
- How zero pricing and reciprocity influence online consumer reviews [Zhang, 2025].
- The role of future time perspective in retirement planning [Polanco, 2025].

These studies provide valuable experimental data and behavioral insights while frequently highlighting their limitations, such as challenges in scaling findings, issues with generalizing results, unexpected effects, and the need for further validation in different contexts. The experimental focus and real-world relevance of this volume make it an excellent resource for applying the GAPMAP framework and the TABI method. These tools can help identify both explicit research gaps, **such as** calls for evidence on the durability of interventions and implicit gaps, such as differences between what people say they believe and how they actually behave, or unresolved factors influencing social preferences. By mapping these gaps, our project aims to uncover new research paths, enhance policy design (like creating more effective nudges or improving financial education), and bridge the gap between controlled experiments and real-world economic behavior.

References

- [1] M. R. Boguslav, N. M. Salem, E. K. White, K. J. Sullivan, M. Bada, T. L. Hernandez, S. M. Leach, and L. E. Hunter, “Creating an ignorance-base: Exploring known unknowns in the scientific literature,” *Journal of Biomedical Informatics*, vol. 143, p. 104405, 2023.
- [2] A. Abd-Alrazaq, A. J. Nashwan, Z. Shah, A. Abujaber, D. Alhuwail, J. Schneider, R. AlSaad, H. Ali, W. Alomoush, A. Ahmed, et al., “Machine Learning–Based Approach for Identifying Research Gaps: COVID-19 as a Case Study,” *JMIR Formative Research*, vol. 8, no. 1, p. e49411, 2024.
- [3] J. Karbach, “Using Toulmin’s model of argumentation,” *Journal of Teaching Writing*, vol. 6, no. 1, pp. 81–92, 1987.
- [4] K. Hyland, “Writing without conviction? Hedging in science research articles,” *Applied Linguistics*, vol. 17, no. 4, pp. 433–454, 1996.
- [5] B. Goujon, “Uncertainty detection for information extraction,” in *Proc. Int. Conf. RANLP*, Borovets, Bulgaria, 2009, pp. 118–122.
- [6] W. W. Chapman, W. Bridewell, P. Hanbury, G. F. Cooper, and B. G. Buchanan, “A simple algorithm for identifying negated findings and diseases in discharge summaries,” *Journal of Biomedical Informatics*, vol. 34, no. 5, pp. 301–310, 2001.
- [7] C. Chen, M. Song, and G. E. Heo, “A scalable and adaptive method for finding semantically equivalent cue words of uncertainty,” *Journal of Informetrics*, vol. 12, no. 1, pp. 158–180, 2018.
- [8] M. R. Boguslav, N. M. Salem, E. K. White, S. M. Leach, and L. E. Hunter, “Identifying and classifying goals for scientific knowledge,” *Bioinformatics Advances*, vol. 1, no. 1, p. vbab012, 2021.
- [9] C. Sutton and A. McCallum, “An introduction to conditional random fields,” *Foundations and Trends® in Machine Learning*, vol. 4, no. 4, pp. 267–373, 2012.
- [10] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[11] A. Bibal, N. M. Salem, R. Cardon, E. K. White, D. E. Acuna, R. Burke, and L. E. Hunter, “RecSOI: recommending research directions using statements of ignorance,” *Journal of Biomedical Semantics*, vol. 15, no. 1, p. 2, 2024.

[12] IPBES, “Knowledge gaps”, Accessed: Aug. 2025. [Online]. Available: <https://www.ipbes.net/knowledge-gaps?page=5>

[13] D. Lahav, J. S. Falcon, B. Kuehl, S. Johnson, S. Parasa, N. Shomron, D. H. Chau, D. Yang, E. Horvitz, D. S. Weld, and T. Hope, “A search engine for discovery of scientific challenges and directions,” in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, no. 13, pp. 11982–11990, 2022.

[14] Elf, M. (Creator), Miedema, E. (Creator), Lipson-Smith, R. (Creator), Kylén, M. (Creator), Saa, J. P. (Creator), Sturge, J. L. (Creator), Nordin, S. (Creator), Bernhardt, J. (Creator), Anåker, A. (Creator) (8 Apr 2025). Database for A Systematic Review of Research Gaps in the Built Environment of Inpatient Healthcare Settings. Zenodo. 10.5281/zenodo.15062860 Available: <https://research.utwente.nl/en/datasets/database-for-a-systematic-review-of-research-gaps-in-the-built-re>

[15] Meta AI, “Llama 3.3 70B Instruct,” Hugging Face, Dec. 6, 2024. [Online]. Available: <https://huggingface.co/meta-llama/Llama-3.3-70B-Instruct>

[16] Meta AI, “Introducing Meta Llama 3.1,” Jul. 23, 2024. [Online]. Available: <https://ai.meta.com/blog/meta-llama-3-1/>

[17] Meta AI, “The Llama 4 herd: The beginning of a new era of natively multimodal intelligence,” Apr. 5, 2025. [Online]. Available: <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>

[18] Google, “Gemma 2 is now available to researchers and developers,” Jun. 27, 2024. Available: <https://huggingface.co/google/gemma-2-9b-it/>

[19] OpenAI, Aug. 7, 2025. Available: <https://openai.com/index/introducing-gpt-5/>

[20] OpenAI, May 13, 2024. Available: <https://openai.com/index/hello-gpt-4o/>

[21] OpenAI, Jul. 18, 2024. Available: <https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/>

- [22] P. Qi, Y. Zhang, Y. Zhang, J. Bolton, and C. D. Manning, “Stanza: A Python natural language processing toolkit for many human languages,” arXiv preprint arXiv:2003.07082, 2020.
- [23] M. Cohen, “Exploring RoBERTa’s theory of mind through textual entailment,” 2021.
- [24] Heuvelink, M. L., Viswanathan, S., Contadini, F. C., Spitters, B., Rappl, C., Cannizzaro, S., ... & Kriegeskorte, N. (2022). THINGS-data, a multimodal collection of large-scale datasets for investigating object representations in human brain and behavior. *eLife*, 12, e82580.
- [25] Eßer, J., Flörchinger, D., Frondel, M., & Sommer, S. (2025). Avoiding cognitive inconsistency: Experimental evidence on sustainable online shopping. *Journal of Behavioral and Experimental Economics*, 119, 102387. <https://doi.org/10.1016/j.socec.2025.102387> (Open access).
- [26] Yokoo, H.-F., Kubo, T., Kunii, D., & Sasaki, H. (2025). When leadership messaging fails to encourage producers’ climate action: Field experiments reveal what works in the real world. *Journal of Behavioral and Experimental Economics*, 119, 102439. (Open access).
- [27] Ocean, N., & Woodman, R. (2025). Unintended consequences of nudges: Challenges in improving attitudes towards shared e-scooters. *Journal of Behavioral and Experimental Economics*, 119, 102437. (Open access).
- [28] Llamosas García, G., & Mazas Pérez-Oleaga, C. (2025). What works in financial education? Experimental evidence on program impact. *Journal of Behavioral and Experimental Economics*, 119, 102401. <https://doi.org/10.1016/j.socec.2025.102401> (Open access).
- [29] Van Campenhout, B., & Nabwire, L. (2025). Buyer-side gender discrimination in bargaining: Evidence from seed sales in Uganda. *Journal of Behavioral and Experimental Economics*, 119, 102404. (Open access).
- [30] Filippin, A., & Pace, N. (2025). The effect of social distancing on trust and solidarity. *Journal of Behavioral and Experimental Economics*, 119, 102402. (Open access).
- [31] Zíka, V., Olšová, P., & Jánská, M. (2025). The attitude–behavior gap is not one-sided: Some do more for the environment than they believe. *Journal of Behavioral and Experimental Economics*, 119, 102446. (Open access).
- [32] Rahal, R.-M. (2025). Advancing openness in economic research through the lens of behavioral and experimental economics. *Journal of Behavioral and Experimental Economics*, 119, 102454. (Open access).

[33] Zhang, X. (2025). Effects of zero price and reciprocity on online consumer reviews. *Journal of Behavioral and Experimental Economics*, 119, 102416. <https://doi.org/10.1016/j.socec.2025.102416>.

[34] Polanco, F. C. (2025). Future time perspective and job satisfaction in financial planning for retirement. *Journal of Behavioral and Experimental Economics*, 119, 102415. <https://doi.org/10.1016/j.socec.2025.102415>.